Mohammad Refat

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Research Interests

Keywords: Computational physics, Scientific computing, Statistical inference, Bayesian inference, Inverse problems, Time-series analysis, Data-driven modeling

Executive Summary: My research focuses on developing computational methods for extracting physical insight from complex data. Across projects in astrophysics, spectroscopy, Galactic archaeology, and time-domain astronomy, I have applied scientific computing, statistical inference, numerical modeling, and inverse methods to study complex physical systems. I am broadly interested in computational physics and in applying these approaches to challenging problems in astrophysics, computational biophysics, and soft condensed matter physics.

Teaching

Adjunct Lecturer Aug 2025 – May 2026
Baruch College
- Introductory Physics Lecture (PHYS 2002) Fall 2025, Spring 2026
- Introductory Physics Laboratory (PHYS 2003) Fall 2025
- Physics II Laboratory (PHYS 3001) Spring 2026

Education

City University of New York (CUNY) Graduate Center 2023–2025
M.S. in Astrophysics
Thesis: *Starspot Inference Using Light Curve Inversion Techniques*

CUNY Baccalaureate for Unique and Interdisciplinary Studies 2019–2022
B.S. in Computational Astrophysics, *cum laude*

Bernard M. Baruch College 2017–2019

Research Experience

Starspot Inference with Light Curve Inversion Techniques Aug 2023 – Sep 2025
American Museum of Natural History
Advisor: Dr. Lucy Lu
- Investigated the feasibility and limitations of reconstructing stellar surface features from rotational photometric variability.
- Applied light-curve inversion, time-series analysis, and Bayesian inference to study non-unique starspot configurations and model degeneracies.
- Developed Python-based computational workflows for simulation, model fitting, and evaluation as part of an M.S. thesis on starspot inference.

Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres May 2021 – Aug 2023
American Museum of Natural History
Advisor: Dr. Johanna Vos

- Investigated atmospheric variability in brown dwarfs and giant exoplanets using rotational photometric observations.
- Applied time-series analysis and statistical modeling to characterize cloud-driven variability and infer atmospheric structure.
- Developed Python workflows for processing observational datasets and analyzing photometric light curves.

Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2 May 2020 – May 2021
Sloan Digital Sky Survey Faculty and Student Team
Advisor: Dr. Allyson Sheffield
- Investigated the chemical and dynamical properties of candidate stellar members of the Jhelum stellar stream using APOGEE-2 spectroscopy.
- Applied statistical analysis to identify stream members and characterize their chemical abundance patterns.
- Contributed to a refereed publication in *The Astrophysical Journal*.

Looking at Star Formation Through Chemistry Aug 2018 – May 2020
American Museum of Natural History
Advisor: Dr. Allyson Sheffield
- Investigated stellar populations through spectroscopic measurements of chemical abundances.
- Processed and normalized echelle spectra and measured equivalent widths of spectral lines.
- Applied quantitative spectroscopic analysis techniques to characterize stellar chemical compositions.

Two-Point Statistics in the Star Forming ISM Jun 2018 – Aug 2018
Flatiron Institute, Center for Computational Astrophysics
Advisor: Dr. Chang-Goo Kim
- Investigated statistical relationships between metallicity, gas dynamics, and star formation in the interstellar medium.
- Applied two-point statistical methods to analyze correlations within numerical simulation data.

Publications

Sheffield, Allyson A., Aidan Z. Subrahimovic, **Mohammad Refat**, et al. (May 2021). "Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2". In: *The Astrophysical Journal* 913.1, p. 39. ISSN: 1538-4357. DOI: 10.3847/1538-4357/abee93. URL: http://dx.doi.org/10.3847/1538-4357/abee93.

Refat, Mohammad (2025). "Starspot Inference Using Light Curve Inversion Techniques". Master's thesis. CUNY Graduate Center. URL: https://academicworks.cuny.edu/gc_etds/6480/.

Academic Presentations

Talks

"Starspot Inference with Light Curve Inversion Techniques" 2025
CUNY Masters Graduation 2025
Flatiron Institute, Center for Computational Astrophysics

"Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres" 2022
240th American Astronomical Society Meeting
American Astronomical Society

"Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres" 2021
Center for Computational Astrophysics / City University of New York / American Museum of Natural History Symposium
Flatiron Institute, Center for Computational Astrophysics

"Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2" 2020
American Museum of Natural History REU Symposium
American Museum of Natural History

"Where Were Stars Born in the Milky Way?" 2019
American Museum of Natural History REU Symposium
American Museum of Natural History

"Two-Point Statistics in the Star Forming ISM" 2018
Center for Computational Astrophysics REU Symposium
Flatiron Institute, Center for Computational Astrophysics

Posters

"Starspot Inference with Light Curve Inversion Techniques" 2025
245th American Astronomical Society Meeting
American Astronomical Society

"Star Spots with Starry" 2024
22nd Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun
University of California San Diego

"Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres" 2023
Center for Computational Astrophysics Gotham Fest 2023
Flatiron Institute, Center for Computational Astrophysics

"Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres" 2023
241st American Astronomical Society Meeting
American Astronomical Society

"Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2" 2021
237th American Astronomical Society Meeting
American Astronomical Society

Technical Skills

Programming	Python, MATLAB, Bash
Scientific Python	NumPy, SciPy, Pandas, Matplotlib, Astropy, Lightkurve, Astroquery
Computational Methods	Time-Series Analysis, Statistical Modeling, Inverse Problems, Bayesian Inference, Computational Modeling, Spectral Analysis
Software	Git, Linux, LaTeX